

AI-Driven Phishing Email Detection Using Natural Language Processing (NLP)

A Machine Learning Approach to Text Classification and Threat Detection

Name: M SRAVAN KUMAR VARMA

CMR Technical Campus, Hyderabad, Telangana, India

Department: Computer Science and Engineering (AIML)

Email: srvnkumr27@gmail.com

Affiliated Institution:

Indian Institute of Computing and Technology (IICT)

Affiliated: I-STEM, Office of the Principal Scientific Adviser to the Government of India

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M Sravan Kumar Varma

Department of Computer Science and Engineering (AI & ML)

CMR Technical Campus, Hyderabad, Telangana, India

srvnkumr27@gmail.com

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Abstract— Phishing emails remain one of the most prevalent vectors for cyberattacks, exploiting linguistic manipulation and social engineering to deceive recipients. This paper presents an end-to-end machine learning pipeline for automated phishing email detection using Natural Language Processing (NLP). A publicly available corpus of 18,650 labeled emails was cleaned, tokenized, and represented using TF-IDF (unigram and bigram) vectorization combined with seven structural metadata features, including URL presence, exclamation frequency, digit count, and uppercase ratio. Four classifiers — Logistic Regression, Random Forest, Multinomial Naive Bayes, and a Multi-Layer Perceptron (MLP) neural network — were trained and benchmarked on an 80/20 stratified split. The MLP classifier achieved the strongest performance with an accuracy of 97.05% and an F1-score of 0.9755, closely followed by Logistic Regression (96.89% accuracy, F1 = 0.9742). Confusion matrix, ROC-AUC, and feature-importance analyses confirm that both textual n-grams and structural metadata contribute meaningfully to detection performance. The results demonstrate that a hybrid text-plus-metadata feature representation, combined with classical and shallow neural classifiers, can deliver production-grade phishing detection accuracy without requiring large-scale deep learning infrastructure.

Index Terms— Phishing detection, Natural Language Processing, TF-IDF, text classification, machine learning, cybersecurity, feature engineering, Random Forest, neural network.

I. INTRODUCTION

Phishing is a form of social engineering attack in which an adversary impersonates a trustworthy entity through email to induce a victim into disclosing sensitive information, transferring funds, or installing malware. Despite advances in spam filtering, phishing emails continue to evade rule-based detectors by mimicking legitimate correspondence and embedding disguised URLs or urgency-based language. Natural Language Processing (NLP) offers a data-driven alternative: by learning statistical patterns in wording, tone, and structure, machine learning models can generalize beyond hand-crafted blacklists to detect novel phishing campaigns.

This work builds a complete phishing-detection system from raw email text to a benchmarked ensemble of classifiers. The objectives are to (i) preprocess and clean a large, noisy corpus of real email text; (ii) engineer both linguistic (TF-IDF n-gram) and structural metadata features; (iii) train and compare four classical and neural classifiers; and (iv) evaluate and interpret results using standard classification metrics, confusion matrices, ROC curves, and feature-importance analysis.

II. RELATED WORK

Prior research on phishing and spam detection has explored rule-based heuristics, blacklist filtering, and, more recently, statistical and deep-learning text classifiers. Bag-of-words and TF-IDF representations paired with classical classifiers such as Naive Bayes and Support Vector Machines have historically formed strong baselines for email classification because email text is high-dimensional and sparse. More recent work incorporates metadata signals — sender domain reputation,

URL structure, and header anomalies — alongside body text, since phishing emails frequently combine linguistic deception with structural red flags. This study follows that hybrid tradition, fusing TF-IDF text features with lightweight structural metadata rather than relying on text alone.

III. METHODOLOGY

A. Dataset

The dataset comprises 18,650 labeled emails sourced from a public phishing-email corpus, each annotated as either "Safe Email" or "Phishing Email." The class distribution is moderately imbalanced, with 11,322 safe emails (60.7%) and 7,328 phishing emails (39.3%), as shown in Fig. 1. The dataset contained 16 records with missing email text, which were removed prior to feature extraction.

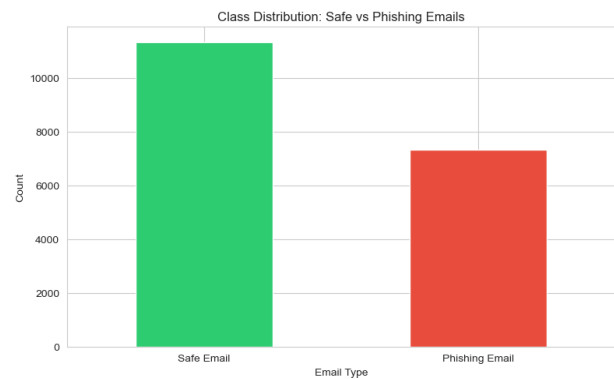


Fig. 1. Class distribution of safe vs. phishing emails in the dataset.

B. Data Preprocessing

Raw email text was cleaned using a deterministic pipeline: HTML tags, embedded URLs, and email addresses were stripped using regular expressions; text was lower-cased; punctuation and numeric digits were removed; and redundant whitespace was collapsed. Records that became empty after cleaning were discarded, leaving 18,629 usable records. This cleaning step ensures that the TF-IDF vectorizer operates on normalized natural-language tokens rather than markup or boilerplate artifacts.

C. Feature Engineering

Two complementary feature families were constructed. First, textual features were derived using a TF-IDF vectorizer configured with unigrams and bigrams, English stop-word removal, a maximum vocabulary of 5,000 terms, and a minimum document frequency of three. Second, seven structural metadata features were engineered directly from the raw (pre-cleaning) text to preserve signals that cleaning would otherwise erase:

- `has_url` — binary indicator of URL presence
- `num_urls` — count of embedded URLs
- `num_exclaim` — count of exclamation marks
- `num_digits` — count of numeric characters
- `text_length` — raw character length
- `num_words` — token count after cleaning
- `uppercase_ratio` — proportion of uppercase characters

Fig. 2 compares the distribution of these metadata features across the two classes, showing that phishing emails exhibit elevated exclamation and uppercase usage relative to safe emails. Metadata features were min-max scaled and horizontally concatenated with the sparse TF-IDF matrix, yielding a combined training matrix of shape $(14,903 \times 5,007)$ and a test matrix of shape $(3,726 \times 5,007)$ after an 80/20 stratified train-test split.

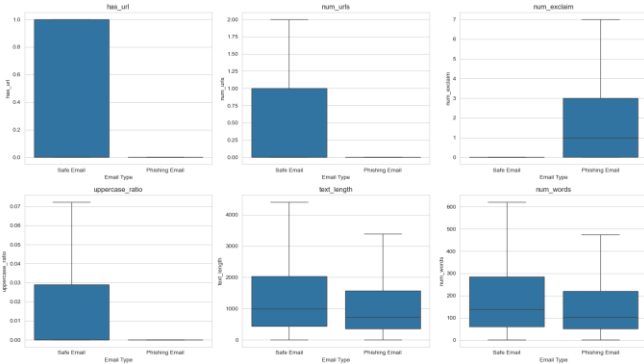


Fig. 2. Distribution of structural metadata features by email class.

D. Model Development

Four classifiers were trained on the combined feature matrix using a fixed random seed for reproducibility: (1) Logistic Regression with L2 regularization; (2) Random Forest with 200 estimators; (3) Multinomial Naive Bayes; and (4) a Multi-Layer Perceptron (MLP) with two hidden layers of 64 and 32 units and

early stopping. Hyperparameter tuning of the best-performing model was subsequently performed using 3-fold grid search cross-validation over regularization strength, tree depth/count, smoothing parameter, and hidden-layer configuration, depending on the model family.

E. Evaluation Metrics

Model performance was assessed using accuracy, precision, recall, and F1-score on the held-out 20% test set, supplemented by confusion matrices, ROC curves with area-under-curve (AUC), and feature-importance analysis to interpret which linguistic and metadata signals drive classification decisions.

IV. RESULTS AND DISCUSSION

A. Model Comparison

Table I summarizes the test-set accuracy and F1-score achieved by each classifier. The MLP neural network achieved the highest performance (accuracy = 97.05%, F1 = 0.9755), narrowly outperforming Logistic Regression (96.89%, F1 = 0.9742) and Random Forest (96.48%, F1 = 0.9706). Multinomial Naive Bayes, while computationally the fastest, trailed the other models with an accuracy of 94.47% and F1-score of 0.9554 — consistent with its simplifying conditional-independence assumption, which is less suited to correlated TF-IDF and metadata features.

Model	Accuracy	F1
Neural Network (MLP)	0.9705	0.9755
Logistic Regression	0.9689	0.9742
Random Forest	0.9648	0.9706
Naive Bayes	0.9447	0.9554

TABLE I. TEST-SET ACCURACY AND F1-SCORE BY MODEL

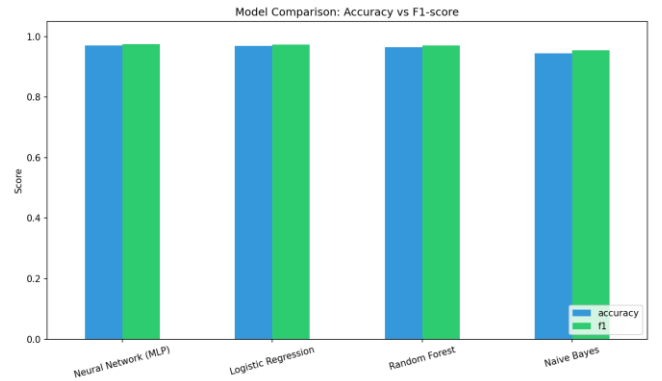


Fig. 3. Accuracy and F1-score comparison across the four classifiers.

B. Confusion Matrix Analysis

Fig. 4 presents the confusion matrices for all four models. All classifiers exhibit a low false-negative rate for phishing emails, which is the operationally critical error type, since a missed phishing email poses greater risk than a false alarm on a safe

email. The MLP and Logistic Regression models show the most balanced error distribution between the two classes.

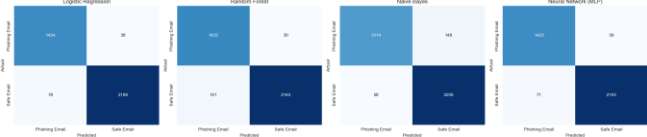


Fig. 4. Confusion matrices for Logistic Regression, Random Forest, Naive Bayes, and MLP.

C. ROC Curve Analysis

Fig. 5 shows the ROC curves for each classifier. All models trace curves close to the top-left corner of the plot, indicating strong separability between phishing and safe emails across decision thresholds, with the MLP and Logistic Regression curves showing the largest area under the curve.

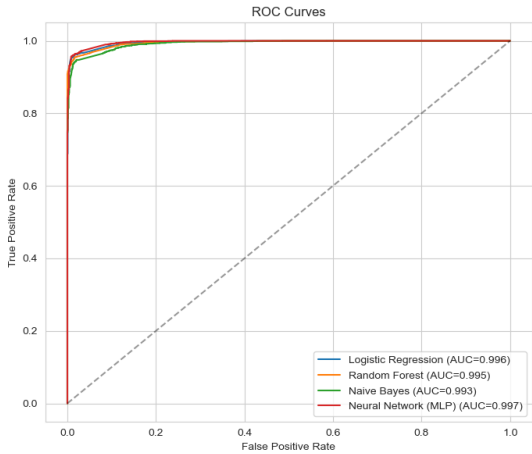


Fig. 5. ROC curves for all trained classifiers.

D. Feature Importance Analysis

To interpret model behavior, the Logistic Regression coefficients (Fig. 6) and Random Forest feature importances (Fig. 7) were examined. The most influential terms for the phishing class are dominated by urgency and monetary language, while safe-email indicators are dominated by conversational and organizational vocabulary. Among the structural metadata features, uppercase ratio and exclamation count contribute the most discriminative signal within the Random Forest model, corroborating the descriptive statistics in Fig. 2.



Fig. 6. Top TF-IDF features driving Logistic Regression predictions, by class.

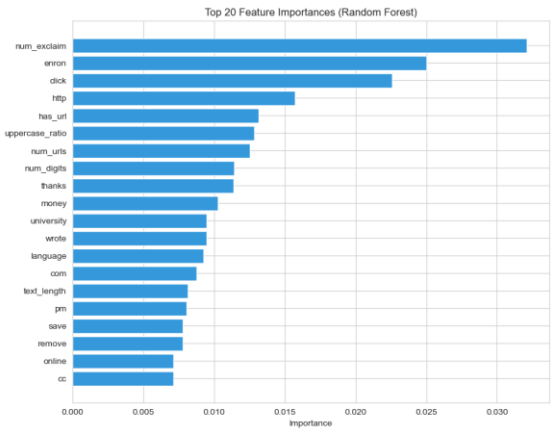


Fig. 7. Top 20 feature importances from the Random Forest classifier.

V. CONCLUSION AND FUTURE WORK

This paper presented a complete phishing-email detection pipeline combining TF-IDF text representation with structural metadata features across four classifiers. The MLP neural network delivered the best overall performance (97.05% accuracy, F1 = 0.9755), demonstrating that a hybrid linguistic-plus-metadata feature set enables high-accuracy phishing detection using comparatively lightweight models. Future work includes incorporating contextual word embeddings (e.g., transformer-based encoders) in place of TF-IDF, extending metadata features to sender-domain reputation and header analysis, and deploying the trained model behind a lightweight web interface for real-time email classification, as outlined in the project's optional deployment phase.

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